Sahaj Bhadja

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Summary

Early-career software engineer with 2+ years building **production AI/LLM applications**, Python and Java services, React interfaces, and scalable APIs; shipped systems handling 10K+ daily requests through hands-on experimentation and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Restructured **fault-tolerant asynchronous LLM workers** using FastAPI, processing 1,000+ jobs with explicit state transitions and failure isolation while tripling throughput under concurrent workloads without cascading task failures.
- Built **Python APIs and relational data services** for an AI platform supporting 500+ users, optimizing indexed queries and access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **full-stack AI grading product** with React, Flask, multimodal LLMs, and OCR, serving 150+ instructors and 10,000+ daily requests while maintaining 90%+ grading consistency in production.
- Integrated **OpenAI, Anthropic Claude, and Qwen providers** behind a switchable architecture, adding OCR fallbacks, token and cost logging, and concurrent processing while reducing feedback turnaround by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a **GPT-based Decision Transformer** for offline RL, improving long-horizon trajectory planning 18% in simulation while developing reproducible data and evaluation pipelines processing 100K+ large-scale trajectory records.
- Built **reproducible PyTorch experiments** with Hydra and Weights & Biases, evaluating model architectures, benchmark results, and failure patterns while improving closed-loop planning stability by 15% across simulation rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **OOP modules with Singleton and Factory patterns** for complex API integrations, optimizing algorithms and AWS Glue transformations to improve response times 30% and reduce technical debt 20%.
- Shipped **cross-platform software through Agile sprints** with Microsoft and GitHub stakeholders, maintaining 20+ unit/integration tests and Jenkins CI/CD across Windows, Linux, and macOS release environments with reliable delivery.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **React and Node.js features** across eight microservices for authentication, billing, resource allocation, and notifications, synchronizing 2M+ records with WebSockets while supporting fast product iteration and demos.
- Automated **Docker packaging, testing, and staged deployments** through Jenkins CI/CD, cutting release time 50% while improving consistency across development environments, code reviews, and reliable production rollout workflows.

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, Bash
AI/LLM: OpenAI API, Anthropic Claude, GPT-4o, Qwen-VL, LLaMA-2, Transformers, LLM Systems, OCR
ML: PyTorch, TensorFlow, NLP, Computer Vision, Feature Engineering, Model Evaluation, Hydra, W&B
Web & Data: React.js, Next.js, Flask, FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB, FAISS
Engineering: Data Structures, Algorithms, OOP, Git, GitHub, Agile/Scrum, Code Review, CI/CD, Docker, PyTest, AWS

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an **LLM document interaction platform** with LLaMA-2, FAISS, Pinecone, and PostgreSQL, indexing 50K+ records for semantic search and summarization while reducing retrieval latency 40% through optimized pipelines.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL development workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript codebase with Docker-backed development workflows.

CaptionCraft Live | *PyTorch, Transformers, Flask, Computer Vision*
- Benchmarked **BLIP, GIT, and ViT-GPT-2 vision-language models** on 10K images and 100K captions, improving caption accuracy from 81% to 96% while building APIs for scalable real-time inference.